Deployment AI Agent + Migration SAS Viya 3.X to SAS Viya 4.X

STEP 1: Plan

- Use the WinScp to transfer cli file to mobaXterm/putty environment.
- Use the putty / mobaXterm

1. Login to Putty or MobaXterm by using IP Address and PPK key (created while deploying SAS Viya on cloudor provided by the admin)
2. Create two directories on the ec2-user level by using this command for inventory Scan:
   - Mkdir -p /inventoryName1/inventoryName2
   - Use command: ls -lt to display the directories and files in the folder
3. Use the copy command to copy the license file to the opt folder by using Sudo privileges.
   - Example: Sudo cp licenseFile /opt
     - License file example: sas-admin-cli-1.18.2-linux-amd64.tgz
   - Move to opt folder by using cd command
     - Cd /opt
   - Use command: ls-lt
   - Change ec2-user to root user by using command: sudo -i
   - Change the permission for License file by using command:
     - tar -xvzf LicenseName
4. Move to "bin" directory by using command: cd bin. Then, use command ls -lt
5. To Install the updated plugins, use command:
   - ./sas-admin plugins
   - Then, use this command to get the list of plugins: ./sas-admin plugins list-repo-plugins
   - After this, use command to install inventory plugin: ./sas-admin plugins install - - repo SAS inventory
   - After this, use command to install Backup plugin: ./sas-admin plugins install - -repo SAS backup
   - After this, use command to install CAS plugin: ./sas-admin plugins install - - repo SAS cas
     - Use the command to check the version of plugins: ./sas-admin "pluginName" - - version
6. Run the command to generate inventory playbook by using "ec2-user". (if does't work use root).
   - ./sas-admin inventory generate playbook -- output-location /tmp/inventory/
   - Follow the link for more clarifications: [SAS Demo | Migration to SAS Viya: Assess and Package the Source 3.x System - YouTube]()
7. Move to /tmp / inventory folder and see all the set of files.
8. Open and edit the credentials.txt file by using command: NANO to edit the file.
   - Change the credentials and save.
   - Copy the file to .sas folder by using command :
   - `cp credentials.txt /home/ec2-user/.sas/`

- change the permission of the credential. txt by using command: chmod 755 file name

9. Before edit viya_plan_backup_vars.yml, create three directories at the level of sas/install.
   - Create three directories by using command: mkdir at the same level.
   -

10. open and edit the viya_plan_backup_vars.yml by using command: nano
    - endpointServer = http://host.example.com
    - customerId:ec2-user
    - deploymentLabel:production
    - inventoryResultsLocation: /sas/install/name1
    - migrationPackageLocation: /sas/install/name2
    - sharedVaultLocation: /sas/install/name3
      - save the file

11. copy the files to the sas/install/ansible/sas_viya_playbook by using commands:
    - cp *.yml /sas/install/ansible/sas_viya_playbook
    - cp *.sh /sas/install/ansible/sas_viya_playbook

12. Copy and change the permission of plugins
    - Copy the plugins to the home/ec2-user/.sas/admin-plugins directory from /sas/admin-plugins directory.
    - Change the permission by using command: chmod 755 fileName
13. Create a sas admin cli profile from **bin** location in root previliages.
    - Create aprofile by using command: ./sas-admin - -k profile init
    - Login by using command: ./sas-admin –k auth login
    - Provide credentials

14. Move to the sas_viya_playbook directory and run the playbook to do the scan.
    - Use command: ansible-playbook viya-plan-backup.yml --tags 'inventoryScan'
15. Move to sas/install/name1 (inventory Result scan) directory and check the files and you can view the report by download and upload .json file to the sas viya.
    - Follow the link to get more help: [SAS Demo | Migration to SAS Viya: Assess and Package the Source 3.x System - YouTube]()

Step 2: Backup

16. To complete the backup, move to the home/sas/install/ansible/sas_viya_playbook directory.
17. To run the backup, use this command: `ansible-playbook viya-plan-backup.yml`
18. Go to migration directory(created in the plan step and mention under the viya_plan_backup_vars.yml file), to find the migration package.

Step 3: Restore

1. Create a CAS custom resource (CR) from each *cas-server-name*.yml file from the SAS Viya 3.X migration package
   - Open the SAS Viya 3.X migration package.
   - Go to the each *cas-server-name* directory in the SAS Viya 3.X migration package by using this command:
     - `cd cas-server-name`
   - Copy the *cas-server-name*.yml file that is in this directory to a known location on the Kubernetes jump box
2. For the further steps use this link [SAS Help Center: Full-System Migration from SAS Viya 3.X: Tasks]
3. Edit the `$deploy/kustomization.yaml.add this command to the resource section:`
   - `- site-config/migration`

4. For restore process follow the link [SAS Help Center: Full-System Migration from SAS Viya 3.X: Tasks]

Step 4: Validation

1. After completing the restore process we need to perform the validation to compare the sas viya 4 and sas viya 3.
2. Follow this link to perform the validation steps :[SAS Help Center: Full-System Migration from SAS Viya 3.X: Tasks]

Step 5: SAS VIYA 4 Installation

1. For deploying sas viya 4, we need to create a cluster in (amazon/AWS) cloud.
   - Cluster name example: "sas-customer-eks"

2.In cluster their will be six ec2 instances.

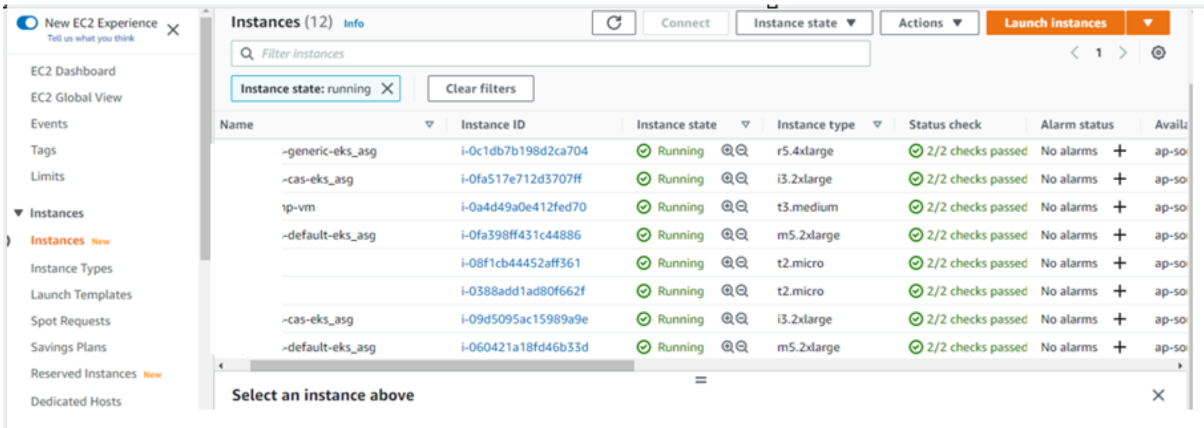
Region: Asia (SYDNEY)
For AWS-EKS , Jump and NFS Servers based on Amazon EC2 On-Demand Pricing
Note: Pricing may vary in the actual cost based on utilization

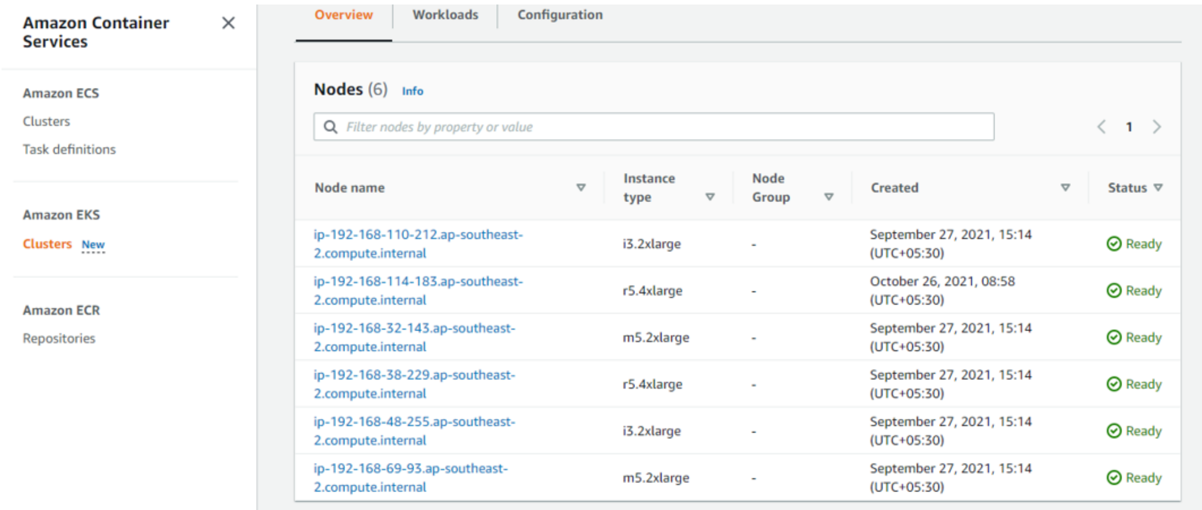
Vendor	Server Name	Server Count	Instance Type	Server Description	Server Version	vCPUs	RAM	Storage Size	Per Hour Rate	Monthly Pricing	Other Notes
AWS	Default node	2	m5.large	AWS EKS		2	8 GiB	EBS Only		80	Up to 10 Gigabit
AWS	Services (Stateful + Stateless + Compute + Connect)	3	m5.4xlarge	AWS EKS		16	64 GiB	EBS Only		60	Up to 10 Gigabit
AWS	CAS	2	i3.2xlarge	AWS EKS		8	61 GiB	1 x 1900 NVMe SSD		12	Up to 10 Gigabit
AWS	Jump Server	1	t3.medium	AWS EC2		2	4 GiB	EBS Only		02	Up to 5 Gigabit
AWS	NFS Server	0	m5.xlarge	AWS EFS		4	16 GiB	EBS Only			Up to 10 Gigabit
AWS	SQL RDS	0	db.m5.xlarge	Internal SAS - Crunchy Postgres		4	16GiB	EBS Only	$		
AWS	EKS Server	1	1 cluster	AWS EKS					$	00	
									TOTAL $	4	USD

SERVERS	URL	CREDENTIALS
	AWS login: https://164705983740.signin.aws.amazon.com/console	USER: user1 PASSWORD: tech123
sas-customer-jump-vm	https://164705983740.signin.aws.amazon.com/console	User name: jumpuser Access key :jumpserver.ppk (ppk file)
sas-customer-eks-generic-eks_asg	https://164705983740.signin.aws.amazon.com/console	User name: jumpuser Access key :jumpserver.ppk (ppk file)
sas-customer-eks-cas-eks_asg	https://164705983740.signin.aws.amazon.com/console	User name: jumpuser Access key :jumpserver.ppk (ppk file)
sas-customer-eks-default-eks_asg	https://164705983740.signin.aws.amazon.com/console	User name: jumpuser Access key :jumpserver.ppk (ppk file)
sas-customer-eks-cas-eks_asg	https://164705983740.signin.aws.amazon.com/console	User name: jumpuser Access key :jumpserver.ppk (ppk file)
sas-customer-eks-default-eks_asg	https://164705983740.signin.aws.amazon.com/console	User name: jumpuser Access key :jumpserver.ppk (ppk file)
sas-customer-eks-default-eks_asg	https://164705983740.signin.aws.amazon.com/console	User name: jumpuser Access key :jumpserver.ppk (ppk file)
sas4-ldap	https://164705983740.signin.aws.amazon.com/console	User name: admin password: CUSTOMERA

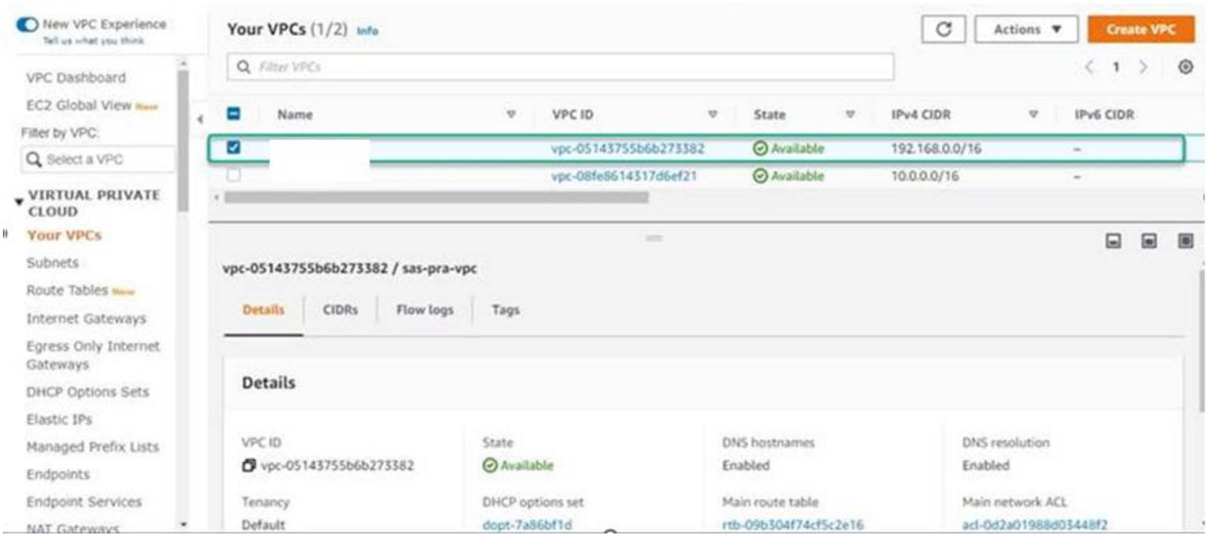
Server names:

Cluster name





VPC Name



3.Create one jump server in cloud (aws/amazon)to communicate with the cluster.

- Jump server name example: sas-customer-jump-vm

4. Added to the terraform script:

Changes to Outputs:

+ autoscaler_account = (known after apply)
+ cluster_endpoint = (known after apply)
+ cluster_name = "sas-customer-eks"
+ cluster_node_pool_mode = "default"
+ cr_endpoint = "https://164705983740.dkr.ecr.ap-southeast-2.amazonaws.com"
+ efs_arn = (known after apply)
+ infra_mode = "standard"
+ jump_admin_username = "jumpuser"
+ jump_private_dns = (known after apply)
+ jump_private_ip = (known after apply)

+ jump_public_dns = (known after apply)
+ jump_public_ip = (known after apply)
+ jump_rwx_filestore_path = "/viya-share"
+ kube_config = (sensitive value)
+ location = "ap-southeast-2"
+ nat_ip = (known after apply)
+ postgres_servers = (sensitive value)
+ prefix = "sas-customer"
+ provider = "aws"
+ rwx_filestore_endpoint = (known after apply)
+ rwx_filestore_id = (known after apply)
+ rwx_filestore_path = "/"
+ worker_iam_role_arn = (known after apply)

5. Cluster created with nodes, EFS(NFS), VPC
   - VPC ID : vpc-098b0ae1968729729
6. Kubernets install
   - Install version of kubectl 1.18.8 or later
   - Use this command: curl -LO https://storage.googleapis.com/kubernetes-release/release/v1.18.20/bin/linux/amd64/kubectl
   - Change the permission by using this command: chmod +x ./kubectl
   - Then, use the root command: sudo mv ./kubectl /snap/bin/kubectl
   - Use command to check the version: kubectl version --short
7. Install Kustomize
   - kustomize 3.7.0
   - Download kustomize here: https://github.com/kubernetes-sigs/kustomize/releases/tag/kustomize%2Fv3.7.0.
   - tar -xvzf 'kustomize_v3.7.0_linux_amd64 (2).tar.gz'
   - cp kustomize /usr/bin
   - kustomize version –short
8. aws install
   - curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
   - apt install unzip
   - unzip awscliv2.zip
9. Aws configure
   - Access key: AKIASMWJ53T6F7APZQUJ
   - Secrete key: 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
   - Eks cluster name: aws eks --region ap-southeast-2 update-kubeconfig --name sas-customer-eks
10. Cert manager
    - kubectl apply -f https://github.com/jetstack/cert-manager/releases/download/v1.5.3/cert-manager.yaml
11. Install ingress
    - Use Kubernetes NGINX Ingress Controller versions 0.41.0 and later
    - Use this command to install helm: snap install helm --classic
    - Use this command : helm repo add ingress-nginx https://kubernetes.github.io/ingress-nginx

- Use this command: helm install nginxingress ingress-nginx/ingress-nginx --set tag=v0.46.0 -n namespace
- Use this command: kubectl --namespace namespacename get services -o wide -w nginxingress-ingress-nginx-controller

12. Install Persistent Storage Volumes, PersistentVolumeClaims, and Storage Classes by using command below:
    - kubectl get storageclass
13. Install pip
    - use this command to install: https://linuxize.com/post/how-to-install-pip-on-ubuntu-18.04/
    - kubectl get nodes -o=custom-columns=NAME:.metadata.name,LABELS:.metadata.labels,TAINTS:.spec.taintskubectl get nodes -o=custom-columns=NAME:.metadata.name,LABELS:.metadata.labels,TAINTS:.spec.taints
    - kubectl label nodes ip-192-168-32-143.ap-southeast-2.compute.internal kubernetes.aws.com/mode=system

- Download deployment assets,certificate and license only from https://my.sas.com/en/my-orders.html

14. Create a directory on the kubectl machine by using command:
    - `mkdir` *`directory-name (create directory- deploy)`*
    - `example Name: deploy`
    - `cd $deploy`
    - `cp examples/security/customer-provided-ingress-certificate.yaml site-config/security`
    - `vi site-config/security/customer-provided-ingress-certificate.yaml`

15. Once your edits are in place, add the path to this file to the generators block of your
    - `$deploy/kustomization.yaml file.`

16. Transformers:
    - `sas-bases/overlays/network/ingress/security/transformers/product-tls-transformers.yaml # adds the tls enablement data bits to selected product DUs`
    - `sas-bases/overlays/network/ingress/security/transformers/backend-tls-transformers.yaml # transformers to support TLS for backend servers`
    - `sas-bases/overlays/required/transformers.yaml # This line is provided as a location reference, so it should appear only once and not be duplicated.`

17. yaml generators:
    - `site-config/security/customer-provided-ingress-certificate.yaml # configures Ingress to use a secret which contains customer-provided certificate and key`

18. export deploy=~/deploy(need to discuss with banupriya)
- cd $deploy

19. mkdir -p site-config/security/cacerts(need to discuss with Banupriya)

20. The following line assumes that your CA Certificates are in a file named /tmp/my_ca_certificates.pem

21. (Need to discuss) cp /tmp/my_ca_certificates.pem site-config/security/cacerts
- cp examples/security/customer-provided-ca-certificates.yaml site-config/security
- vi site-config/security/customer-provided-ca-certificates.yaml

22.Yaml generators:
- site-config/security/customer-provided-ca-certificates.yaml # generates a configmap that contains CA Certificates

22. kustomize build . | kubectl -n sasviyaoperator apply -f –

23. Created directory license for certificate and license file:

- Created directory deploy for deployment - kustomization.yaml
- Site-config folder with storageclass.yaml,postgres and security with required changes.

24. Docker install
- Use this command: apt install docker.io
- cat sas-bases/examples/kubernetes-tools/password.txt | docker login cr.sas.com --username '9CLK78' --password-stdin
- docker pull cr.sas.com/viya-4-x64_oci_linux_2-docker/sas-orchestration:1.41.6-20210513.1620868672630
- docker logout cr.sas.com
- docker tag cr.sas.com/viya-4-x64_oci_linux_2-docker/sas-orchestration:1.41.6-20210513.1620868672630 sas-orchestration

25. kustomize build -o site.yaml

26. collecting containers information use this command:
- docker run --rm -v $(pwd):/sasviya/ sas-orchestration create sas-deployment-cr --deployment-data /sasviya/license/'SASViyaV4_9CLK78_certs (1).zip' --license /sasviya/license/'SASViyaV4_9CLK78_1_lts_2021.1_license_2021-09-24T041301 (1).jwt' --user-content /sasviya/deploy --cadence-name lts --cadence-version 2021.1 > sasviya-sasdeployment.yaml

27. Use this command: kubectl -n sasviyaoperator apply -f sasviya-sasdeployment.yaml

28. Use the command:kubectl -n sasviyaoperator get sasdeployment

29. Use the command: kubectl get jobs -n sasviyaoperator -l "sas.com/backup-job-type=migration" -L "sas.com/sas-backup-id,sas.com/backup-job-type,sas.com/sas-restore-status"

30. Provide DNS Name:

- Example: DNS Name - prodsas4.customeraustralia.com.au

31. Results of the latest readiness check:

- kubectl wait -n sasoperator --for=condition=ready pod --selector="app.kubernetes.io/name=sas-readiness" --timeout=1800s

32. On the sas viya 4 login page(https://prodsas4.customergroup.com.au/SASDrive/)Sign in as user: sasboot,password:CUSTOMERSAS

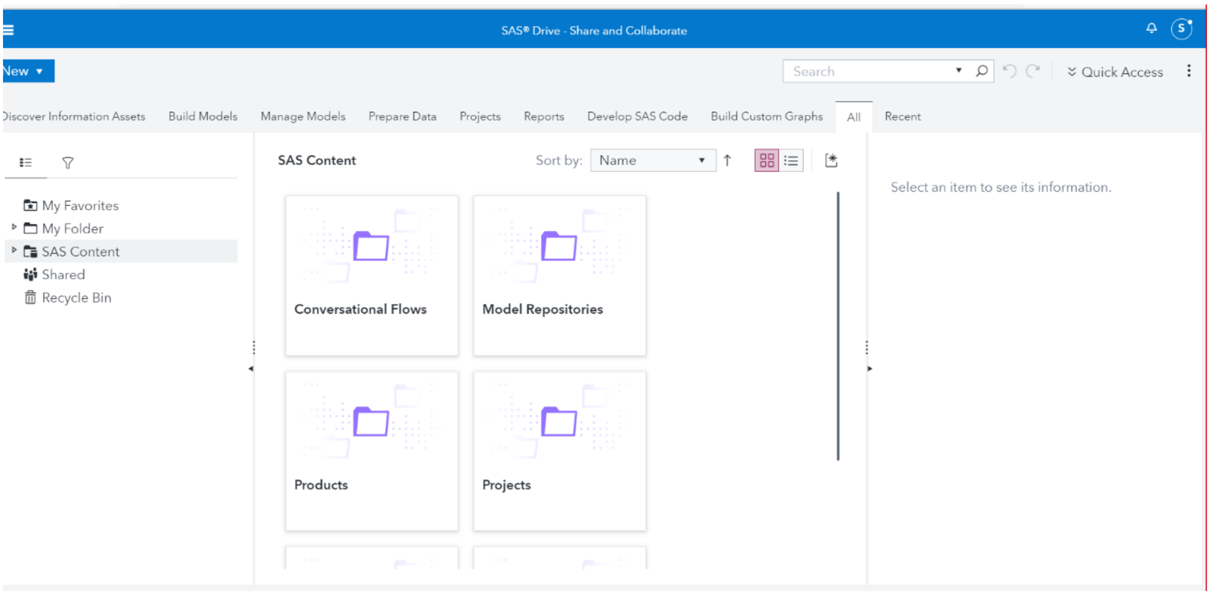
33. Searching command: kubectl logs -n sasoperator -c sas-logon-app $(kubectl -n sasoperator get pods | grep sas-logon | cut -f1 -d' ') | grep sasboot

*******34. https://prodsas4.customeraustralia.com.au/SASLogon/reset_password?code=BbL3jBX8Gv

35.Delete commands: kubectl delete secrets -n sasoperator --selector "sas.com/deployment=sas-viya"

36. Delete command kubectl delete clusterroles --selector "app.kubernetes.io/name=sas-deployment-operator"

37. https://prodsas4.customergroup.com.au/SASLogon/reset_password?code=ZKXPY9h7vs



38.
Extracting the SAS Viya generated root CA Certificate from a Deployment

If you make use of cert-manager generated certificates to protect SAS Viya servers and services, the SAS Viya deployment process creates a Kubernetes secret named "sas-viya-ca-certificate-secret" that contains the root CA certificate used to issue the necessary server identity certificates. This happens when you

* Use full-stack TLS mode
* Use front-door TLS mode and choose to configure the NGINX Ingresses to use a cert-manager certificate
In these configured modes, the cert-manager root CA certificate will need to be added to the bundle of trusted root CA certificates used by any external clients that will access the SAS Viya deployment. These clients include:
*Web browsers
*SAS Viya Administrative CLI applications
*Remote SAS sessions that connect to Viya
*curl

39. The following command can be used to extract the CA certificate from the secret. You will need to run this with an Kubernetes identity that has roles which allow it to access secrets.

```text
#command to get the secret
kubectl get secret sas-viya-ca-certificate-secret -o go-template='{{(index .data "ca.crt")}}'
```

Because Kubernetes stores secrets in base64 encoding, you must decode the certificate before it can be used.

```text

40. #command to get the secret, decode it into PEM format and write it to a file
kubectl get secret sas-viya-ca-certificate-secret -o go-template='{{(index .data "ca.crt")}}' | base64 -d > /tmp/my_ca_certificate.pem

41.
https://saseksaws.apps.Techtechnology.com.au//SASLogon/reset_password?code=r9OgKxa4lX

42. kubectl annotate ingress kubernetes.io/ingress.class=nginx --all -n sasviyaoperator

43. kubectl logs -n sasviyaoperator -c sas-logon-app $(kubectl -n sasviyaoperator get pods | grep sas-logon | cut -f1 -d' ') | grep sasboot

44. kubectl get jobs -n sasviyaoperator -l "sas.com/backup-job-type=migration" -L "sas.com/sas-backup-id,sas.com/backup-job-type,sas.com/sas-restore-status"

45. NFS mounting:

❖ https://docs.aws.amazon.com/efs/latest/ug/installing-amazon-efs-utils.html

❖ sudo mount -t efs fs-73607d4b nfs/
/home/jumpuser/nfs/migration

❖ https://prodsas4.customeraustralia.com.au/SASLogon/reset_password?code=V8LZrEanz

46.Certificate installation:

❖ https://askubuntu.com/questions/645818/how-to-install-certificates-for-command-line

❖ Script for whole deployment:

❖ https://communities.sas.com/t5/Administration-and-Deployment/VIYA4-Deploy-Viya-4-Anywhere-Tutorial-Script-Locally-or-in-cloud/td-p/779037

47. LDAP Configuration:

❖ https://computingforgeeks.com/install-and-configure-openldap-server-ubuntu/
❖ https://www.digitalocean.com/community/tutorials/how-to-install-and-configure-openldap-and-phpldapadmin-on-an-ubuntu-14-04-server
https://computingforgeeks.com/how-to-configure-ubuntu-as-ldap-client/

❖ prodsas4.customergroup.com.au
```

- `Customeraus2021 - ldap password`
- `sudo dpkg-reconfigure slapd`

48. uninstall LDAP:

- https://installlion.com/ubuntu/xenial/main/s/slapd/uninstall/index.html

49. /etc/apache2

50. Standard ports are 389 for ldap and 636 for ldaps.

51. User Credentials

Site	Username/Password
https://prodsas.customeraustralia.com.au/	username: SASAdmin Password: CUSTOMERAus2020
https://prodsas4.customergroup.com.au/	Username: sasboot Password: CUSTOMERAus2021
https://164705983740.signin.aws.amazon.com/console	username: user1 Password: password
https://prodsas4.customeraustralia.com.au/SASStudio/	sasboot password Techtech123
OPEN LDAP URL	http://13.211.168.227/lam/templates/login.php Password: CUSTOMERAus2021

52. Validation report steps:

we have completed validation report as per sas document mentioned below Link, and other steps followed Fyi

https://documentation.sas.com/doc/en/sasadmincdc/v_017/calmigration3x/p0s8n6d5si7oqun1ixkvuw10mzcx.htm#p1i6iynmof3andn1c2w9zoyreg8h

-Create Validate Reports – This task has been completed

1. Use this command to Create a profile if you have not already and log in as a SAS administrator

sas-viya --profile profile-name profile init

sas-viya --profile profile-name auth login

Note:

profile-name=prod

2. Use this command to Store the SAS administrator credentials to be used to perform the ETL using the compute plug-in to the sas-viya CLI

sas-viya --profile profile-name compute credentials create

Note:

profile-name=prod

3. Navigate to the $deploy/sas-bases/examples/backup directory.

cd sasviya/deploy/sas-bases/examples/backup

4. Copy the $deploy/sas-bases/examples/backup directory to the $deploy/site-config/backup directory

cp -r ~/sasviya/deploy/sas-bases/examples/backup ~/sasviya/deploy/site-config/backup

5. Make sure that the Execute permission is set on the generate-backup-job script:

chmod +x generate-backup-job.sh

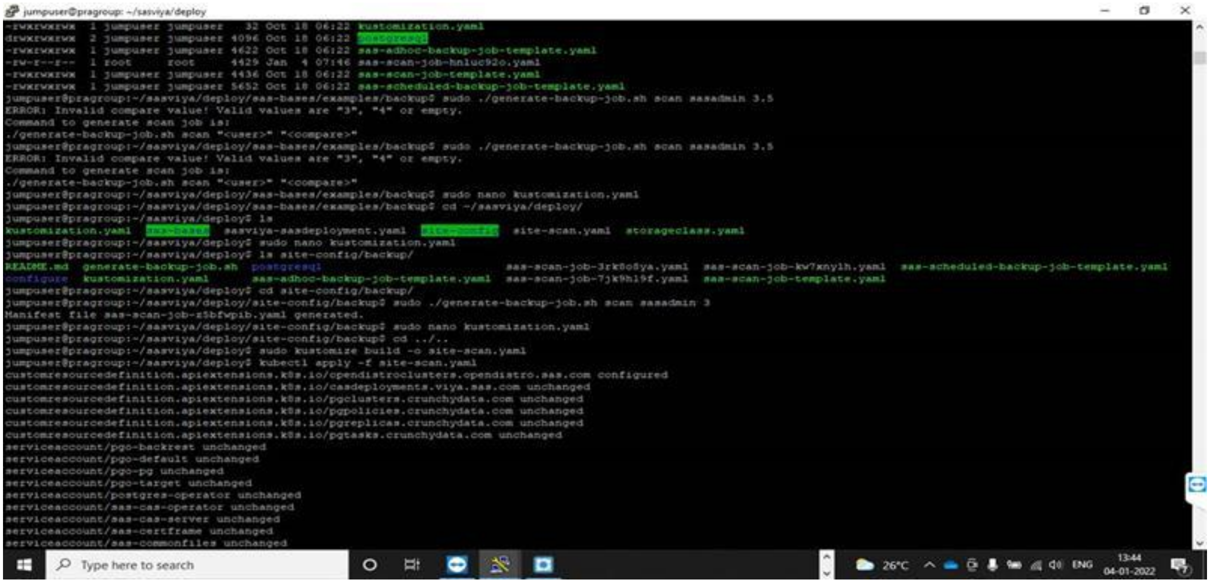
6. Make sure that Write permission is set on the kustomization.yaml file.

chmod +w kustomization.yaml

7. Execute the script generate-backup-job.sh to generate the manifest file for the Kubernetes scan job.

./generate-backup-job.sh scan User compare

where User is sasadmin and compare is 3



8. Edit the kustomization.yaml file. Remove any previous entries from the resources section. Then add the name of the YAML file generated in the previous step.

resources:

  sas-scan-job-*xxx*.yaml

9. Navigate to the $deploy directory. Edit the kustomization.yaml file and add the following to the resources section:

resources:

  site-config/backup

10. Build the manifest using the following command:

kustomize build -o site-scan.yaml

11. Create the inventory scan job by applying the manifest:

kubectl apply -f site-scan.yaml

SAS Help Center
documentation.sas.com